

Algebra 1 – Q1.4_1.7 C

Name: _____

1.4 – Fractions, Decimals & Percents

1. I can convert between fractions, decimals, and percents

Convert as directed. Write answers in simplest form.

- a. $\frac{2}{5} \rightarrow$ decimal: _____ percent: _____
- b. $0.375 \rightarrow$ fraction: _____ percent: _____
- c. $12\% \rightarrow$ fraction: _____ decimal: _____
- d. $2.4 \rightarrow$ fraction: _____ percent: _____

2. I can decide if a fraction's decimal ends or repeats (and write it).

For each fraction: **write the decimal** (use a bar for repeating digits).

- a. $\frac{7}{16} \rightarrow$ decimal: _____
- b. $\frac{8}{11} \rightarrow$ decimal: _____
- c. $\frac{5}{6} \rightarrow$ decimal: _____

3. I can compare and order fractions, decimals, and percents in real-world situations

Three stores are discounting jackets originally priced at **\$120**:

- Store A: **\$20 off**
- Store B: **15% off**
- Store C: **" $\frac{1}{4}$ off"**

Which store gives the **biggest discount**? _____

1.5 – Multiply, Divide & Compare Fractions

4. I can multiply fractions and mixed numbers

Simplify your answers.

a. $\frac{2}{3} \times 9 =$ _____

b. $\frac{7}{10} \times \frac{5}{7} =$ _____

c. $3\frac{1}{2} \times 1\frac{1}{4} =$ _____

5. I can divide fractions by using reciprocals

Find the following quotients.

a. $\frac{4}{5} \div \frac{2}{3} =$ _____

b. $9 \div \frac{4}{7} =$ _____

1.6 – Solving Problems with Fractions, Decimals & Percents

6. I can find a part, a whole, or a percent

- a. What is 65% of 200? _____
- b. 21 is 30% of what number? _____
- c. What percent is 35 out of 56? _____

7. I can use percents to solve problems involving grades, sales, and survey results

- a. An \$85 backpack is 20% off. After the discount, 9% sales tax is added. What is the final price? (Round to the nearest cent.)

Answer: _____

- b. In a survey of 250 students, 52% prefer drama. How many students is that? _____

8. I can solve percent change problems in real-world contexts

A recipe uses $1\frac{1}{2}$ cups of sugar. You **reduce** the sugar by 30%. How much sugar do you use now?

Answer: _____ cup(s)

1.7 – Order of Operations (PEMDAS)

9. I follow the correct order of operations (including fractions and negatives)

Simplify each expression.

a. $10 - \frac{5}{4} \cdot 4 =$ _____

b. $-2^3 + 9 =$ _____

c. $\frac{1}{3} + \frac{3}{5} \cdot 5 - 2 =$ _____

d. $\frac{9}{4} \div \left(\frac{3}{8}\right) - 2 =$ _____

10. I avoid common mistakes when simplifying expressions

A student says $8 \div \frac{2}{5} = 4$ “because you divide by 2.”

a. Explain what is wrong.

b. Find the correct value of $8 \div \frac{2}{5}$.

11. I avoid common mistakes with negatives and exponents

True or False? Explain your reasoning clearly.

“ $(-5)^2 = 25$ ”